
TraceAV-Bench: Benchmarking Multi-Hop Trajectory Reasoning over Long Audio-Visual Videos

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<https://heinz217.github.io/TraceAV-Bench-Page>

Abstract

Real-world audio-visual understanding requires chaining evidence that is sparse, temporally dispersed, and split across the visual and auditory streams, whereas existing benchmarks largely fail to evaluate this capability. They restrict videos to short clips, isolate modalities, or reduce questions to one-hop perception. We introduce **TraceAV-Bench**, the first benchmark to jointly evaluate *multi-hop reasoning over long audio-visual trajectories* and *multimodal hallucination robustness*. TraceAV-Bench comprises 2,200 rigorously validated multiple-choice questions over 578 long videos, totaling 339.5 hours, spanning 4 evaluation dimensions and 15 sub-tasks. Each question is grounded in an explicit reasoning chain that averages 3.68 hops across a 15.1-minute temporal span. The dataset is built by a three-step semi-automated pipeline followed by a strict quality assurance process. Evaluation of multiple representative OmniLLMs on TraceAV-Bench reveals that the benchmark poses a persistent challenge across all models, with the strongest closed-source model (Gemini 3.1 Pro) reaching only 68.29% on general tasks, and the best open-source model (Ming-Flash-Omni-2.0) reaching 51.70%, leaving substantial headroom. Moreover, we find that robustness to multimodal hallucination is largely decoupled from general multimodal reasoning performance. We anticipate that TraceAV-Bench will stimulate further research toward OmniLLMs that can reason coherently and faithfully over long-form audio-visual content.

1 Introduction

The rapid advancement of Multimodal Large Language Models (MLLMs) [1–6] has broadened the perceptual horizon of language models, enabling them to process visual [7, 8] and auditory [9, 10] information far beyond the boundaries of language alone. Building upon this foundation, Omnimodal Large Language Models (OmniLLMs) [11–17] have recently emerged, designed to jointly perceive and reason over real-world text, vision, and audio within a unified framework, achieving competitive performance across tasks such as audio-visual understanding and cross-modal reasoning [18–20]. This capability is fundamental for interacting with the physical world in a human-like manner, in which information from different modalities is simultaneously integrated to form coherent understanding.

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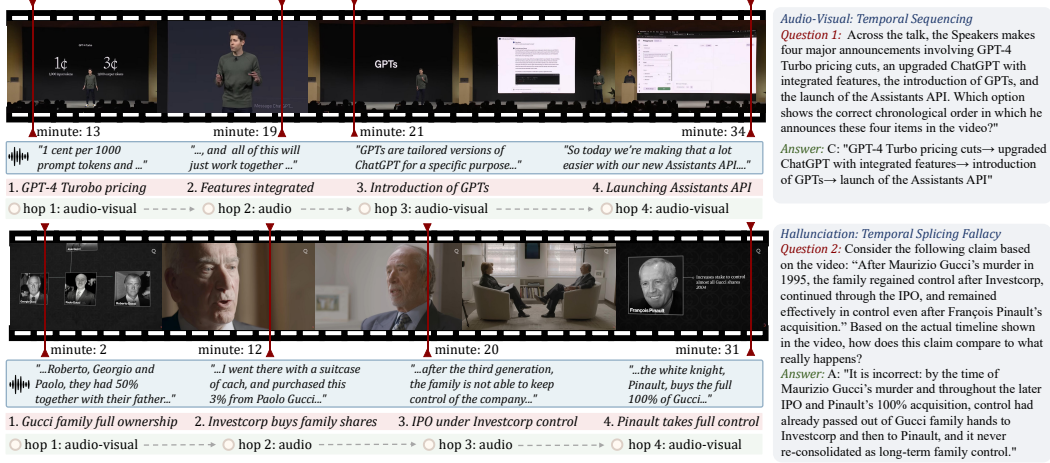


Figure 1: Illustrative examples from TraceAV-Bench, each grounded in an explicit multi-hop evidence trajectory whose hops are tagged with their source modality. The first example (*Temporal Sequencing*, AVR) requires chronologically ordering four events by chaining speech and on-screen cues; the second (*Temporal Splicing Fallacy*, MH) requires rejecting a fabricated narrative that splices temporally isolated events into a false ownership timeline. Both showcase the main idea of TraceAV-Bench: composing evidence from temporally dispersed, cross-modal clues over long videos.

However, despite these promising results, bringing OmniLLMs into genuine real-world utility reveals several unresolved challenges. 1) **Long-form multi-hop trajectory reasoning** poses the most critical bottleneck [21–24]. While OmniLLMs can handle simple, short-clip perception tasks, real-world scenarios often demand chaining clues scattered across tens of minutes of continuous audio-visual content. This requires reasoning from a *trajectory* of temporally dispersed evidence spanning multiple events, rather than a single moment. 2) **Cross-modal information fusion** remains equally unresolved [25, 26]. A model that can process audio and visual streams does not necessarily know how to jointly leverage their complementary information under complex conditions, particularly when evidence in one modality is only interpretable in the context of the other. 3) **Multimodal hallucination** further compounds these difficulties [27–29]. When jointly processing mixed audio-visual inputs, OmniLLMs sometimes generate responses that are inconsistent with the input or not grounded in the observed content, raising serious reliability concerns. Critically, these challenges remain not only largely unsolved, but also insufficiently studied. Existing benchmarks mostly operate on short video clips of at most a few minutes, evaluate visual and audio modalities in isolation, or restrict questions to shallow single-hop inference, failing to comprehensively evaluate these challenges [30, 31].

To this end, we introduce **TraceAV-Bench**, a comprehensive benchmark designed to evaluate OmniLLMs on multi-hop reasoning over long audio-visual trajectories. TraceAV-Bench is built upon 578 long videos totaling 339.5 hours, spanning diverse genres and multiple languages, and it is, to the best of our knowledge, the *first* benchmark to simultaneously require multi-hop trajectory reasoning and assess multimodal hallucination robustness in the context of long audio-visual content. Via a three-step, semi-automated data synthesis pipeline, we construct 2,200 rigorously validated multiple-choice questions (MCQs) across 4 evaluation dimensions and 15 sub-tasks, covering **Audio-Visual Joint Reasoning**, **Visual-Centric Reasoning**, **Audio-Centric Reasoning**, and **Multimodal Hallucination**. Every question in TraceAV-Bench is grounded in an explicitly annotated multi-hop evidence trajectory (as showcased in Figure 1), where each reasoning chain spans at least two temporally non-adjacent events, and averages 3.68 hops across a temporal span of 15.1 minutes.

Our main contributions can be summarized as follows:

- **The TraceAV-Bench Benchmark.** We present a large-scale, carefully curated benchmark that extends the evaluation frontier for OmniLLMs by demanding genuine cross-modal, multi-hop trajectory reasoning and hallucination robustness over long audio-visual content, providing the research community with a tool to rigorously diagnose model capabilities beyond existing benchmarks.
- **A Scalable Semi-Automated Construction Pipeline.** We propose a three-step pipeline that decouples visual captioning, asynchronous audio-visual fusion, and LLM-based question generation,

Table 1: Comparison of TraceAV-Bench with existing audio-visual benchmarks. V, I, A: video, image, audio. Anno.: M = manual, A&M = automatic + manual.

Benchmark	Modality	Duration	Avg. Length	Multi-hop	Multi-domain	Hallucination	Anno.	#Q
<i>Short-clip benchmarks (≤ 60 s)</i>								
AVQA [32]	V+A	~ 10 s	~ 10 s	×	×	×	M	57,335
Music-AVQA [33]	V+A	~ 60 s	~ 60 s	×	×	×	M	45,867
AVTRUSTBENCH [34]	V+A	10–60 s	-	×	✓	✓	A&M	600 K
AV-Odyssey [31]	I+A	/	/	×	✓	×	M	4,555
AVHBench [28]	V+A	~ 10 s	~ 10 s	×	✓	✓	A&M	5,302
OmniBench [30]	I+A	-	-	×	✓	×	M	1,142
Daily-Omni [25]	V+A	30–60 s	42.8 s	×	×	×	A&M	1,197
<i>Long-form video benchmarks (> 60 s)</i>								
WorldSense [35]	V+A	15–656 s	141.1 s	×	✓	×	M	3,172
JointAVBench [36]	V+A	-	97.2 s	×	✓	×	A&M	2,853
OmniVideoBench [26]	V+A	4–1955 s	384.2 s	×	✓	×	M	1,500
LVOmniBench [21]	V+A	613–5482 s	2,070 s	×	✓	×	M	1,014
TraceAV-Bench (Ours)	V+A	606–8394 s (Ultra-long)	2,112 s	✓	✓	✓	A&M	2,200

enabling the construction of logically coherent, trajectory-grounded MCQs at scale. A rigorous multi-stage quality assurance process further ensures that every retained question is of high quality.

- **Comprehensive Empirical Analysis with Key Findings.** We benchmark state-of-the-art models spanning open-source and closed-source OmniLLMs and MLLMs. Our results reveal that all models struggle significantly on TraceAV-Bench, with even the strongest model achieving only 68.29% accuracy. Further analysis of model-specific failure patterns illuminates the limitations of current OmniLLMs and provides insights into future model development.

2 Related Work

2.1 Omnimodal Large Language Models

Real-world perception is inherently multi-sensory, and closing the gap between human and machine understanding requires models that can handle vision, audio, and language. The research community has broadened the perceptual scope of LLMs, giving rise to Vision-Language Models (VLMs) [1, 11, 37–41], Audio-Language Models (ALMs) [42–46], and more recently OmniLLMs [11, 47–52] capable of processing text, images, videos, and audio within a single framework. Both open-source models [12, 13, 15, 53–55] and closed-source models such as the Gemini series [17], which combine multimodal pre-training with large context windows, have demonstrated strong performance across audio and visual tasks. As these capabilities mature, models are increasingly positioned to handle extended audio-visual inputs. However, rigorous evaluation in this regime remains scarce.

2.2 MLLM Benchmarks

MLLM benchmarks have evolved with model capabilities, progressing from modality-isolated suites for image [56–58] and audio [59–62] understanding to specialized axes such as fine-grained perception [63–66] and multimodal reasoning [67–76]. The rise of OmniLLMs has spurred audio-visual benchmarks [24, 25, 30, 31, 77–79], including a few targeting long-form videos [21, 26, 80, 81], but their questions are typically answerable from a single moment, leaving *multi-hop reasoning over long audio-visual trajectories* and *multimodal hallucination robustness* in long-form settings unexplored. As summarized in Table 1, TraceAV-Bench is the first to require *multi-hop reasoning* across temporally distant evidence spans, a property absent in all existing audio-visual benchmarks, alongside a comprehensive hallucination evaluation dimension.

2.3 Dataset Collection

To evaluate the multi-hop reasoning capabilities of OmniLLMs over long trajectories in an audio-visual context, we define eligible long videos as those exceeding **10 minutes**, possessing semantically diverse and rich scene transitions, and containing aligned, meaningful information in both visual and audio modalities.

Video Collection and Deduplication. Our initial video corpus is mainly sourced from three publicly available video understanding benchmarks: OmniVideoBench [26], LVBench [22] and

VideoMME [23], where YouTube is their primary source, there is a natural risk of data overlap. To ensure the uniqueness of our benchmark, we apply a rigorous deduplication process based on video metadata and content hashing, yielding a diverse initial pool of candidate videos.

Video Quality Filtering. We further process the candidate videos through a strict filtering pipeline with the following criteria with a half-automated manner:

- **Visual Dynamics and Event Density:** Videos must exhibit sufficient visual dynamics, with frequent scene transitions and a steady stream of distinguishable events. Therefore, videos with fewer than 3 scene transitions are directly discarded.
- **Modality Completeness:** TraceAV-Bench explicitly evaluates the joint understanding of audio and visual streams. Thus, videos without an audio track or with a mean volume below -50 dB (indicating silence or negligible background noise) are discarded.
- **Multi-hop Reasoning Potential:** Our annotators manually review the videos to assess their potential for extracting multi-hop questions. Specifically, videos are retained only if they exhibit high intrinsic complexity, such as interwoven story lines, multiple interacting entities, linked events, and coherent narrative structures. Importantly, these elements must feature distinct cross-modal dependencies scattered across long temporal trajectories.

3 TraceAV-Bench

After executing this filtering process, we finalized a high-quality collection of **578** videos, yielding a total of **339.5** hours of continuous audio-visual content ready for complex multi-hop construction. As shown in Figure 2, the retained videos span a wide range of genres, ensuring broad topical diversity. Notably, the corpus is also multilingual, covering English, Chinese, and other major languages.



Figure 2: Video category distribution.

3.1 Task Definition

To comprehensively evaluate multi-hop reasoning capabilities of OmniLLMs over long trajectories in long videos, we design 15 distinct sub-tasks (with detailed definitions in Appendix A). These tasks are categorized into four core evaluation dimensions, briefly described as follows:

- **Audio-Visual Joint Reasoning (AVR):** This category forms the core of TraceAV-Bench, evaluating the model’s ability to chain clues across both visual and auditory streams over long temporal spans. It includes seven sub-tasks: *Information Retrieval (IR)*, *Temporal Sequencing (TS)*, *Entity Tracking (ET)*, *Forward Causal Reasoning (FCR)*, *Backward Causal Reasoning (BCR)*, *Cross-Modality Matching (CMM)*, and *Spatiotemporal Localization (SL)*.
- **Visual-Centric Reasoning (VR):** Rather than joint reasoning, this dimension isolates the visual modality to evaluate how well the model maintains its visual reasoning capabilities when faced with long video inputs. It comprises *Spatial Reasoning (SR)* and *Visual Counting (VC)*.
- **Audio-Centric Reasoning (AR):** Complementary to VR, this dimension aims to assess the model’s auditory capabilities under long-video settings. It encompasses examination across three distinct aspects: *Speech Context (SC)*, *Environmental Sound (ES)*, and *Background Music (BM)*.
- **Multimodal Hallucination (MH):** This category specifically investigates whether the model exhibits hallucinations in audio-visual contexts. It tests model robustness through *Visual-to-Audio Deception (V2A)*, *Audio-to-Visual Deception (A2V)*, and *Temporal Splicing Fallacy (TSF)*.

3.2 Data Construction Pipeline

Synthesizing high-quality multi-hop questions over long videos is a significant challenge. Purely manual construction is prohibitively expensive at this scale, while end-to-end generation from current OmniLLMs is limited by context windows, modality bias, and cross-modal hallucinations when

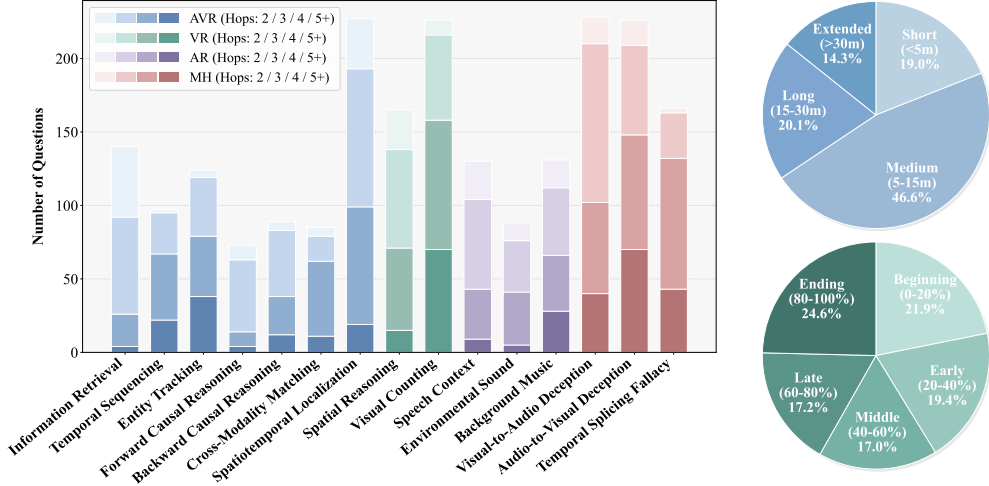


Figure 3: Statistics analysis of TraceAV-Bench. *left*: per-sub-task question counts by evaluation dimension (AVR, VR, AR, MH), stacked by hop count (2/3/4/5+). *right top*: distribution of video durations. *right bottom*: distribution of the position for all evidence hops along the video timeline.

processing ultra-long audio-visual streams. To address these limitations and ensure the logical depth of TraceAV-Bench, we propose a decoupled three-step semi-automated construction pipeline. This pipeline distills long videos into dense textual event catalogs that ground question generation.

Step 1: Minute-Level Visual Captioning. To address context window limits in ultra-long videos, we first segment each video into 1-minute clips. A key design is to process only the visual modality in isolation, as prior work demonstrates that concurrent audio-visual input can introduce cross-modal hallucinations [28, 82]. We deploy Qwen3-VL-32B-Instruct [1] to process each clip sequentially and extract fine-grained captions. A critical mechanism in this step is the *Entity Cache*: at each minute, the model receives the accumulated dictionary of all previously identified entities, including persons, objects, and locations, identifies those present, registers newly discovered ones, and produces a highly descriptive visual summary. This guarantees the consistency of the captions produced.

Step 2: Audio-Visual Fusion. Building on the visual narrative from Step 1, we adopt an *asynchronous fusion* strategy aligning the audio stream with existing visual descriptions. The corresponding 1-minute audio segments are processed by Gemini-2.5-Flash [17], prompted to condition on both the visual narrative and entity library produced in Step 1. The model has two objectives: (1) to synthesize a unified audio-visual narrative integrating speech, sounds, and music with the visual description; and (2) to perform an entity update, selectively updating entity descriptions based on new audio evidence. This yields a chronologically aligned, high-density bimodal event catalog for question generation.

Step 3: Agentic Question Generation. To reduce the cognitive load of multi-hop question generation over ultra-long videos, we decompose this step into a three-stage agentic workflow powered by GPT-5.1 [83], structuring the catalogs into MCQs. In the **first stage**, an *Event Segmentation Agent* abstracts the sequence of minute-level captions into higher-level event blocks, because adjacent minutes often belong to the same scene and exposing an LLM to dozens of minutes introduces information overload. The agent therefore scans the sequence with a three-minute context window, assessing whether the current minute continues the event or marks a boundary, and merging contiguous minutes accordingly. Each resulting event block carries a fused narrative summary and retains the fine-grained minute-level annotations from previous steps. In the **second stage**, a *Trajectory Proposal Agent* operates on event blocks and selects candidate reasoning chains, where each trajectory must span at least two temporally non-adjacent blocks, sharing a clear causal or logical entity link. Proposed trajectories are filtered by a minimum temporal gap constraint to prevent trivial single-hop constructions. In the **third stage**, a *Question Generation Agent* converts each validated trajectory into a four-option MCQ under three strict and hierarchical generation principles:

- **Anti-Shortcut Formulation:** The question stem must not be answerable via general world knowledge or common sense. It must require direct observation of the content named in the trajectory.
- **Trajectory-Grounded Distractors:** All incorrect options must be semantically plausible and grounded in actual events from the same video, preventing elimination by surface-level dissimilarity.

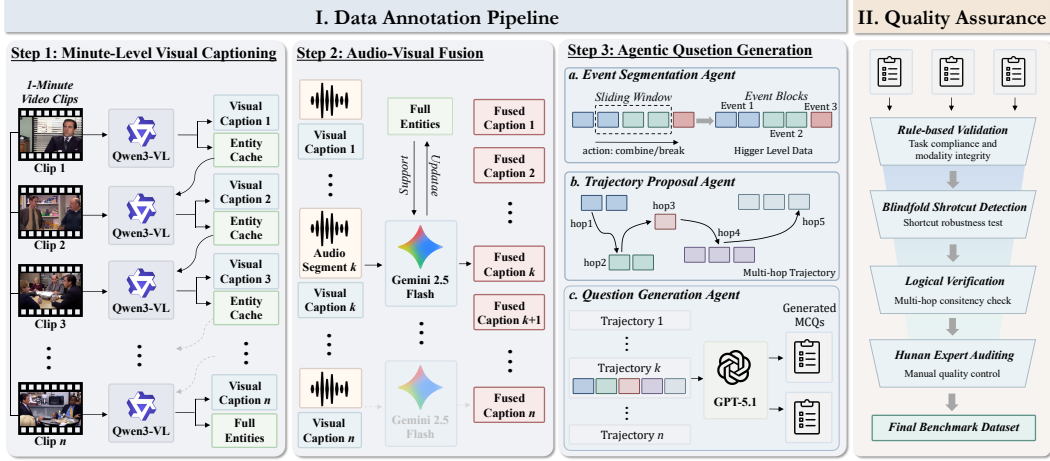


Figure 4: Overview of the TraceAV-Bench data construction pipeline.

- **Stylistic Uniformity:** All options must be comparable in length, syntactic structure, and wording style to prevent guessing via surface-level cues.

3.3 Quality Assurance

Despite the strict generation constraints applied in Step 3, the model can still produce questions with latent flaws. To ensure the collected questions are high-quality and genuinely multi-hop, we process the generated candidates through a rigorous quality assurance stage led by the following principles:

- **Rule-based Validation:** We verify that all questions strictly adhere to their predefined tasks. For example, we enforce that AVR tasks genuinely necessitate evidence from both modalities and that unimodal tasks contain no cross-modal leakage. We additionally verify that each evidence hop is anchored to a specific and verifiable minute-level timestamp and that the MCQ format is strictly adhered to. Questions that fail any of these checks are automatically discarded.
- **Logical Verification:** A separate LLM conducts a deep audit of each surviving item’s multi-hop integrity. It explicitly filters out “pseudo multi-hop” questions, checks for answer leakage within the question stem, and ensures that all distractors are semantically relevant and plausibly confusable, avoiding obvious stylistic or length disparities.
- **Blindfold Shortcut Detection:** Even with anti-shortcut prompting during generation, some questions may remain answerable through linguistic biases or world knowledge without viewing the video. We therefore submit each item to Gemini 2 Flash with only the question stem and options as text input, without all video content, and record whether the model selects the correct answer. Any item answered correctly by this “blindfolded” solver is removed from the pool.
- **Human Expert Auditing:** As a final quality gate, human annotators manually review a stratified 15% random sample drawn from the automatically filtered pool. Reviewers replace unclear or weak answer options, remove questions that are too similar to others within the same video, adjust the difficulty as needed, and verify that the reasoning paths require multi-hop integration across the referenced evidence. We also enforce a strict batch-rejection policy: if the error or ambiguity rate within any sampled batch exceeds 5%, the entire batch is permanently discarded.

3.4 TraceAV-Bench Statistics

Video statistics. The benchmark is built upon **578** long videos with a total duration of **339.5 hours**. Video lengths range from 10.1 to 139.9 minutes, with a mean of 35.2 minutes. The majority (57.6%) fall in the 30–60 minute range, and 43 videos (7.4%) exceed one hour, underscoring the ultra-long-form nature of the benchmark. All videos feature resolutions up to 4K, with 73.7% at HD (720p) or above, and each contains a stereo audio track encompassing speech, sounds, and music.

Question statistics. TraceAV-Bench contains **2,200** questions across four dimensions: Audio-Visual Joint Reasoning (AVR, 7 tasks, 835 questions, 38.0%), Visual-Centric Reasoning (VR, 2 tasks, 391 questions, 17.8%), Audio-Centric Reasoning (AR, 3 tasks, 349 questions, 15.9%), and Multimodal

Table 2: Main results on TraceAV-Bench (%) across 12 general sub-tasks. Sub-tasks grouped by three dimensions: **AVR** (*Audio-Visual Joint Reasoning*), **VR** (*Visual-Centric Reasoning*), **AR** (*Audio-Centric Reasoning*). Abbreviations: IR = *Information Retrieval*, TS = *Temporal Sequencing*, ET = *Entity Tracking*, FCR = *Forward Causal Reasoning*, BCR = *Backward Causal Reasoning*, CMM = *Cross-Modality Matching*, SL = *Spatiotemporal Localization*, SR = *Spatial Reasoning*, VC = *Visual Counting*, SC = *Speech Context*, ES = *Environmental Sound*, BM = *Background Music*. **Avg** is the average over all tasks. Best result per task is **bolded**, and best open-source result is underlined.

Model	AVR							VR		AR			Avg
	IR	TS	ET	FCR	BCR	CMM	SL	SR	VC	SC	ES	BM	
Closed-source OmniLLMs (With Visual and Audio)													
Gemini 3.1 Pro	83.57	60.82	71.77	86.30	61.80	49.41	51.54	73.94	41.15	96.92	63.64	78.63	68.29
Gemini 2.5 Pro	83.57	63.92	60.48	76.71	50.56	54.12	48.02	68.48	39.38	83.85	65.91	67.18	63.52
Gemini 3 Flash	82.14	53.61	65.32	83.56	59.55	49.41	29.07	73.33	36.73	86.92	61.36	66.41	62.28
Gemini 2.5 Flash	75.00	58.76	62.10	75.34	58.43	40.00	29.07	66.06	39.38	81.54	60.23	62.60	59.04
Gemini 2 Flash	66.43	53.61	58.06	64.38	43.82	41.18	25.11	54.55	27.43	70.00	55.68	58.78	51.59
Open-source OmniLLMs (With Visual and Audio)													
Ming-Flash-Omni-2.0	56.43	53.61	47.58	57.53	40.45	44.71	31.28	65.45	39.38	63.85	56.82	63.36	51.70
Qwen3-Omni-30B-A3B	47.14	51.55	35.48	43.84	50.56	40.00	32.60	58.18	38.50	63.85	59.09	60.31	48.43
OmniVinci-9B	49.29	44.33	38.71	57.53	34.83	35.29	33.48	55.15	34.51	65.38	65.91	54.20	47.38
MiniCPM-o 4.5	45.71	36.08	37.90	28.77	26.97	41.18	37.44	60.61	38.50	61.54	59.09	64.12	44.83
Qwen2.5-Omni-7B	46.43	32.99	37.10	30.14	35.96	37.65	37.44	49.70	35.40	60.00	52.27	49.62	42.06
Gemma 4-E4B	39.29	38.14	37.10	36.99	29.21	36.47	16.74	55.15	34.07	55.38	54.55	58.02	40.93
Video-SALMONN 2	42.14	41.24	29.03	30.14	29.21	32.94	31.28	48.48	39.38	47.69	47.73	44.27	38.63
HumanOmni-7B	37.86	31.96	29.84	31.51	31.46	25.88	35.68	55.15	34.96	52.31	51.14	44.27	38.50
VideoLLaMA2.1-AV-7B	36.43	29.90	25.81	17.81	16.85	25.88	22.91	38.79	38.94	36.15	39.77	35.88	30.43
Baichuan-Omni-1.5	37.14	14.43	20.97	15.07	12.36	20.00	30.40	37.58	26.99	32.31	42.05	33.59	26.91
Open-source Single-Modality MLLMs													
Qwen3-VL-32B	44.29	39.18	32.26	38.36	38.20	34.12	16.30	67.27	39.38	46.15	48.86	48.09	41.04
Qwen3-VL-8B	34.29	28.87	29.84	26.03	24.72	24.71	17.62	59.39	32.30	45.38	42.05	46.56	34.31
Qwen2-Audio-7B	30.71	27.84	33.06	20.55	26.97	29.41	29.96	26.67	23.45	38.46	37.50	44.27	30.74
Open-source OmniLLMs (Visual Only)													
Ming-Flash-Omni-2.0	37.86	35.05	32.26	45.21	31.46	27.06	19.82	54.55	38.94	43.08	47.73	49.62	38.55
Qwen3-Omni-30B-A3B	37.86	34.02	37.90	32.88	34.83	24.71	30.84	53.94	38.94	38.46	40.91	43.51	37.40

Hallucination (MH, 3 tasks, 625 questions, 28.4%). The benchmark comprises 1,848 single-choice (84.0%) and 352 multi-choice questions (16.0%).

Multi-hop trajectory statistics. Each question is grounded in a multi-hop reasoning chain spanning specific minutes of the video. On average, each question requires **3.68** evidence hops, with a temporal span averaging **15.1 minutes** across the trajectory.

4 Experiments

4.1 Experimental Settings

Evaluated Models. We benchmark models in four categories: **1) Closed-source OmniLLMs:** Gemini 2, Gemini 2.5 [17] and Gemini 3; **2) Open-source OmniLLMs:** Qwen3-Omni [54], Qwen2.5-Omni [55], OmniVinci [53], MiniCPM-o [84], HumanOmni [13], Video-SALMONN 2 [15], Baichuan-Omni-1.5 [16], VideoLLaMA2.1 [85], Ming-Flash-Omni [12], and Gemma 4; **3) Single-modality MLLMs:** video-only Qwen3-VL [1] and audio-only Qwen2-Audio [45]; **4) Visual-only ablations:** Ming-Flash-Omni and Qwen3-Omni with the audio stream removed.

Evaluation Protocol. For open-source models, we follow official inference configurations and sample as many frames as permitted by the context window to maximize performance. For closed-source models, we use the recommended sampling rate (1 frame per second). A response is correct only if the predicted option(s) exactly match the ground-truth answer(s).

Table 3: Hallucination robustness on the MH dimension of TraceAV-Bench (%). V2A = *Visual-to-Audio Deception*, A2V = *Audio-to-Visual Deception*, TSF = *Temporal Splicing Fallacy*. **MH Avg** is the average over the three MH sub-tasks. **Gen. Avg** is the average over the 12 general sub-tasks (Table 2). Models are sorted by MH Avg within each group. **Rank (MH→Gen.)** shows each model’s within-group rank by MH Avg and by Gen. Avg respectively. Best result per task is **bolded**, and best open-source result is underlined.

Model	V2A	A2V	TSF	MH Avg	Gen. Avg	Rank (MH→Gen.)
<i>Closed-source OmniLLMs</i>						
Gemini 3.1 Pro	89.57	79.91	84.34	84.61	68.29	1 → 1
Gemini 3 Flash	76.52	75.55	87.35	79.81	62.28	2 → 3
Gemini 2 Flash	74.78	81.66	77.11	77.85	51.59	3 → 5
Gemini 2.5 Pro	79.13	74.24	75.90	76.42	63.52	4 → 2
Gemini 2.5 Flash	60.87	66.81	66.87	64.85	59.04	5 → 4
<i>Open-source OmniLLMs</i>						
Qwen3-Omni-30B-A3B	65.65	69.87	66.87	<u>67.46</u>	48.43	1 → 2
Ming-Flash-Omni-2.0	71.30	67.25	62.65	67.07	<u>51.70</u>	2 → 1
Gemma 4-E4B	<u>74.35</u>	69.43	56.02	66.60	40.93	3 → 6
MiniCPM-o 4.5	70.87	<u>72.05</u>	56.63	66.52	44.83	4 → 4
Qwen2.5-Omni-7B	60.43	55.46	53.61	56.50	42.06	5 → 5
OmniVinci-9B	42.17	44.10	42.17	42.81	47.38	6 → 3
Video-SALMONN 2	45.65	39.30	37.95	40.97	38.63	7 → 7
HumanOmni-7B	33.91	37.99	28.31	33.40	38.50	8 → 8
Baichuan-Omni-1.5	28.26	41.48	22.29	30.68	26.91	9 → 10
VideoLLaMA2.1-AV-7B	35.22	29.69	19.88	28.26	30.43	10 → 9

4.2 Quantitative Results

Overall Performance. Table 2 reports per-task accuracy for 12 general sub-tasks across the AVR, VR, and AR. The results show that TraceAV-Bench poses a genuine challenge for all evaluated models. The strongest model, Gemini 3.1 Pro, achieves only 68.29% accuracy. Closed-source Gemini series models demonstrate *dominant performance*, consistently occupying the top of the rankings, with Ming-Flash-Omni-2.0 (51.70%) being the best-performing open-source model, narrowly trailing Gemini 2 Flash. The overall experimental results suggest that *the ability of models to execute logical hops between audio and visual information across long videos still poses a significant challenge*.

Audio-Visual Joint Reasoning. Within the AVR dimension, heterogeneous sub-task performance suggests that joint audio-visual reasoning is *a multifaceted rather than a monolithic capability*. Even models with extended context windows reveal that context capacity alone does not translate uniformly into multi-hop reasoning ability. While they excel at straightforward, perception-oriented tasks (e.g., IR task), they encounter bottlenecks in *logical synthesis and causal inversion over long durations*. Models widely regarded for strong audio-visual understanding, despite their extended context windows that in principle favor long-video comprehension, still exhibit limitations in multi-hop reasoning ability. Specifically, models demonstrate relative competence in straightforward, perception-oriented tasks (e.g., IR), but encounter significant bottlenecks when performing higher-order reasoning that requires causal inversion or the synthesis of logical dependencies. This is most evident in the consistent FCR/BCR asymmetry across all model groups. For instance, Gemini 3.1 Pro (FCR 86.30%, BCR 61.80%) and Ming-Flash-Omni-2.0 (FCR 57.53%, BCR 40.45%) show substantially lower BCR performance, which requires tracing an observed effect to its cause across temporally dispersed evidence, than FCR, which asks models to predict the outcome of a given cause.

Visual-Centric and Audio-Centric Reasoning. For VR tasks, Visual counting has long posed a challenge for multimodal models, and this holds equally in our setting. All models show substantially lower performance on VC than on SR. Regarding AR tasks, audio processing capabilities vary substantially across architectures. The Gemini series maintains a dominant lead in auditory understanding, while among open-source models, OmniVinci achieves competitive audio understanding despite its relatively smaller parameter count. Across all models, SC scores are consistently higher than ES and BM, revealing that the models are relatively *stronger at processing speech information than at interpreting environmental sound and music*.

Hallucination Robustness. Table 3 reports model accuracy on three MH sub-tasks: *Visual-to-Audio Deception* (V2A), where fabricated audio cues contradict verified visual evidence; *Audio-to-Visual*

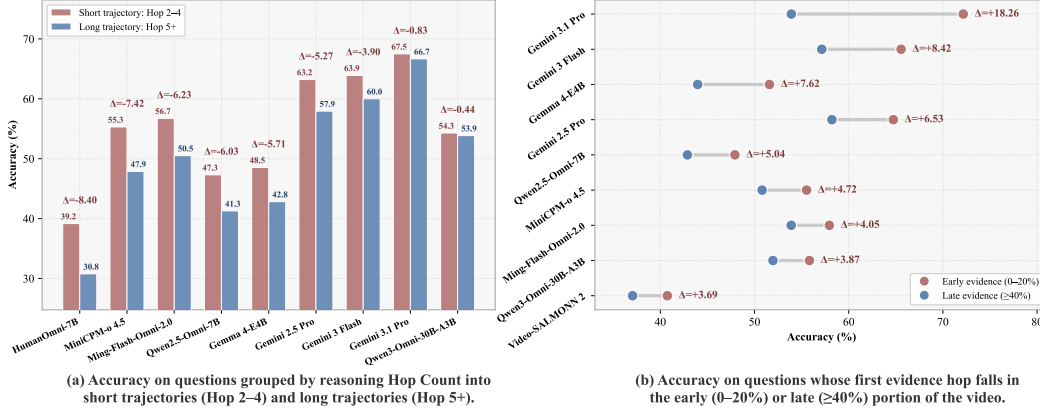


Figure 5: Two diagnostic analyses of OmniLLMs on TraceAV-Bench.

Deception (A2V), where misleading audio cues contradict verified visual content; and *Temporal Splicing Fallacy (TSF)*, where temporally isolated fragments are falsely presented as a coherent causal sequence. The central observation is that **hallucination robustness and general-task ability are largely decoupled**. Gemma 4 is the most striking example, ranking 3rd on MH Avg (66.60%) while only 6th on Gen. Avg (40.93%). OmniVinci shows the inverse, ranking 3rd on Gen. Avg (48.28%) but 6th on MH Avg (42.82%). Across groups, leading open-source models such as Qwen3-Omni-30B-A3B and Ming-Flash-Omni-2.0 attain reasonably strong hallucination robustness, with a gap to frontier closed-source models still remaining. Moreover, models exhibit heterogeneous behavior across hallucination types, suggesting that more fine-grained evaluation and targeted adjustments are essential to train models with reduced hallucination.

4.3 Diagnostic Analyses Beyond Aggregate Accuracy

Here we re-slice the predictions along two structural axes that the benchmark explicitly controls, namely the length of the evidence trajectory and the position of its first hop within the video, and surface a number of findings.

Long reasoning trajectories sharply degrade performance. Figure 5(a) groups questions by hop numbers and contrasts short chains (Hop 2–4) with long chains (Hop 5+). Most models degrade consistently as trajectory lengthens, with MiniCPM-o 4.5 losing 7.42% and Ming-Flash-Omni-2.0 losing 6.23%. The Gemini family is more resilient thanks to their larger context windows, with Gemini 3.1 Pro losing only 0.83% and Gemini 3 Flash 3.90%, while still exhibiting a clear downward trend. Since enlarging context alone offers only limited relief, this calls for training data that explicitly supplies multi-hop audio-visual evidence chains, encouraging a “collect-then-aggregate” reasoning pattern rather than simply relying on parametric memorization.

Models prefer the opening of a video over its middle and tail. Figure 5(b) compares accuracy for questions whose first evidence hop falls in the early portion (0–20%) with those whose first hop appears in the middle or late portion ($\geq 40\%$). The gap is *positive for every evaluated model*, ranging from +3.69% (Video-SALMONN 2) to +18.26% (Gemini 3.1 Pro), with a mean of +6.91%. Strikingly, scaling does not mitigate this preference, as the strongest closed-source model shows the large gap, suggesting larger models amplify rather than correct this tendency to over-rely on early segments. OmniLLMs thus treat the early portion as a privileged anchoring region while failing to localize later cues, calling for new frame sampling strategies that go beyond uniform sampling and reallocate the perceptual budget toward middle and tail segments.

5 Conclusion

In this paper, we present TraceAV-Bench, the first benchmark to jointly stress *multi-hop trajectory reasoning* and *multimodal hallucination robustness* in the long audio-visual video context. Built upon a decoupled three-step semi-automated pipeline and a rigorous multi-stage quality assurance process, TraceAV-Bench comprises 2,200 trajectory-grounded multiple-choice questions over 578

long videos, where every question is anchored in an explicit multi-hop evidence chain spanning both modalities. We comprehensively evaluate representative OmniLLMs across the general and hallucination dimensions of TraceAV-Bench, and our analyses surface several insightful findings that we believe offer meaningful guidance for future model training and evaluation. We hope TraceAV-Bench serves as a rigorous testbed for the next generation of OmniLLMs that reason coherently and faithfully over long, evidence-sparse audio-visual content.

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A Detailed Task Definitions

This section gives the full definition of each of the fifteen sub-tasks of TraceAV-Bench, grouped by the four evaluation dimensions introduced in the paper. For every sub-task we specify both *what the question asks* and, more importantly, *which capability of OmniLLMs the task is designed to probe*.

A.1 Audio-Visual Joint Reasoning (AVR)

The AVR dimension is the conceptual core of TraceAV-Bench. Every AVR sub-task is constructed so that the answer cannot be recovered from any single modality nor from any single moment in the video. Instead, the model must assemble evidence that is simultaneously dispersed across the visual and auditory channels and along the long temporal axis. The seven sub-tasks below probe orthogonal axes of cross-modal multi-hop reasoning.

- **Information Retrieval (IR):** IR probes whether a model can retrieve a specific fact (an on-screen number, a spoken name, an embedded text) when the retrieval requires fusing partial cues from at least two distant events, for instance, when the identity of an entity is established at one moment and a fact about that entity is uttered much later. The task therefore stresses long-context cross-modal grounding rather than simply fact recall.
- **Temporal Sequencing (TS):** Even when models recover the right set of facts, they often lose the timeline. TS asks the model to chronologically order a small set of events or entity-state changes drawn from non-adjacent positions in the video, isolating the question of whether internal representations preserve temporal order across long spans.
- **Entity Tracking (ET):** A persistent challenge for OmniLLMs is re-identifying the same entity after the visual surface changes (clothing, lighting, viewpoint) or the auditory surface changes (different speaker, ambient noise). ET targets the capability that the model must follow a single entity across multiple separate events and reason about its evolving role, relationships, or state.
- **Forward Causal Reasoning (FCR):** FCR begins with an early “cause” event and asks for its eventual consequence in a much later event. The reasoning chain must traverse intermediate events whose direct relevance is not immediately obvious, isolating the model’s ability to project consequences forward across a long timeline.
- **Backward Causal Reasoning (BCR):** BCR operates in the reverse direction of FCR. The question is anchored to a late observable outcome and asks for its earlier root cause. We deliberately distinguish between forward and backward causality because the two directions exercise different inference patterns. Backward reasoning, in particular, requires the model to distinguish the true cause from spurious antecedents that merely co-occur along the timeline.
- **Cross-Modality Matching (CMM):** CMM probes audio-visual binding under temporal dispersion. The model must pair an auditory cue (a distinctive voice, a sound effect, a musical motif) with its correct visual referent among several visually plausible candidates that appear at different times. The task is designed to expose models that handle each modality competently in isolation but fail to bind them when the evidence is temporally separated.
- **Spatiotemporal Localization (SL):** SL asks for the minute-level timestamp at which a specified target event occurs, where the timestamp itself must be triangulated by chaining clues from at least two different events. By forcing the model to commit to an explicit time, the task externalizes whether the model carries a faithful internal clock.

A.2 Visual-Centric Reasoning (VR)

The VR dimension isolates the visual modality by drawing all evidence from the visual stream alone. Its two sub-tasks correspond to the two visual capabilities most commonly degraded by long context:

- **Spatial Reasoning (SR):** SR asks about spatial relationships (relative positions, directions, layouts) that must hold consistently across multiple events. A correct answer requires the model to maintain a stable mental map of the scene over many minutes of video, rather than re-derive geometry from a single sampled frame.
- **Visual Counting (VC):** Open-ended counting is a well-known failure mode of MLLMs. VC further conditions the count on a constraint that is itself defined by another event (e.g., “*how many times does action X occur after event Y begins?*”), so that the counting target cannot even be located without first solving an additional reasoning hop.

A.3 Audio-Centric Reasoning (AR)

The AR dimension complements VR by isolating the auditory channel. Because audio in long videos arrives as a continuous stream rather than as discrete chunks, the three sub-tasks each target a distinct acoustic sub-stream: speech, non-speech environmental sound, and background music.

- **Speech Context (SC):** SC requires reasoning over spoken dialogue or narration spanning multiple events. To preempt visual leakage, the questions are constructed so that the answer is fully derivable from what is said and not from what is seen, providing a controlled probe of long-form speech comprehension that the visual stream cannot shortcut.

Algorithm 1: Agentic Question Generation Pipeline (Step 3).

Input : Bimodal event catalog $\mathcal{M} = [m_1, \dots, m_N]$; task type τ ; span threshold δ .

Output : Released MCQ set $\mathcal{Q}$.

Stage 1 – Event Segmentation Agent:

```
 $\mathcal{E} \leftarrow \emptyset$ ; open event  $e \leftarrow \{\text{captions: } [m_1], \text{ summary: } m_1\}$ 
for  $t = 2$  to  $N$  do
   $(a, s') \leftarrow \text{SegAgent}(e.\text{summary}, m_t, m_{t+1}, m_{t+2})$ 
  if  $a = \text{CONTINUE}$  then
    extend  $e$  with  $m_t$ ;  $e.\text{summary} \leftarrow s'$ 
  else
     $\mathcal{E} \leftarrow \mathcal{E} \cup \{e\}$ ; open new event from  $m_t$ 
 $\mathcal{E} \leftarrow \mathcal{E} \cup \{e\}$ 
```

Stage 2 – Trajectory Proposal Agent:

```
 $\mathcal{C} \leftarrow \text{ProposeAgent}(\mathcal{E}, \tau)$ 
 $\mathcal{T} \leftarrow \{c \in \mathcal{C} : \text{span}(c) \geq \delta \wedge \text{adjacent events of } c \text{ share a named entity}\}$ 
```

Stage 3 – Question Generation Agent:

```
 $\mathcal{Q} \leftarrow \emptyset$ 
for each trajectory  $t \in \mathcal{T}$  do
   $q \leftarrow \text{QGenAgent}(t, \tau)$ 
  refine each per-hop timestamp against the raw minute captions
  randomly permute the four answer slots of  $q$ 
   $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{q\}$ 
```

return $\mathcal{Q}$

- **Environmental Sound (ES):** ES targets non-speech, non-musical sounds such as a siren, a closing door, or machinery. These cues are typically brief and easily missed, and threading them across distant events tests whether the model genuinely attends to acoustic events.
- **Background Music (BM):** BM asks about background music or ambient soundscape, including how musical shifts align with the unfolding narrative. The task probes a form of long-horizon audio understanding that carries strong narrative signal in real-world long videos.

A.4 Multimodal Hallucination (MH)

The MH dimension departs from accuracy-style evaluation and instead measures whether a model can refuse a fabricated premise. Long audio-visual content is a fertile ground for hallucination, because evidence is sparse and the model must commit to an answer under uncertainty. A model that aces AVR but cannot reject false premises is not yet trustworthy.

- **Visual-to-Audio Deception (V2A):** The question is grounded in genuine visual evidence drawn from the video and then asks about an audio detail that does not occur anywhere in the soundtrack. The intended response is to reject the audio premise rather than confabulate one, which exposes models that over-rely on visual grounding to “fill in” the missing audio half.
- **Audio-to-Visual Deception (A2V):** A2V is the dual of V2A, where the question is anchored in genuine audio evidence and asks about a visual detail that never actually appears. This direction specifically targets models with strong language priors that synthesize visual content consistent with the spoken context, even when no such visual content was ever observed.
- **Temporal Splicing Fallacy (TSF):** TSF presents a fabricated narrative that splices real but temporally isolated fragments into a single ostensibly coherent sequence. The model must explicitly identify the chronological inconsistency, which directly tests whether the model reasons about the global temporal layout of evidence rather than verifying each fragment in isolation.

Table 4: Overview statistics of TraceAV-Bench, organized along three axes: the source video corpus, the released question pool, and the per-question multi-hop reasoning trajectories.

Video Corpus		Question Pool		Reasoning Trajectories	
Total videos	578	Total questions	2,200	Evaluation dimensions	4
Total duration	339.5 hrs	Single-choice	1,848 (84.0%)	Sub-tasks	15
Duration range	10.1–139.9 min	Multi-choice	352 (16.0%)	Avg. hop count	3.68
Avg. duration	35.2 min	Avg. question length	43.6 words	Avg. temporal span	15.1 min
Resolution	73.7% HD or above	Avg. option length	22.1 words	Modality coverage	audio + visual

B More details of the Agentic Question Generation Pipeline

Algorithm 1 consumes the per-minute bimodal catalog $\mathcal{M}$ produced by Steps 1 and 2 and a single span threshold δ that controls how temporally distant the evidence in a multi-hop chain must be, and it returns the set of trajectory-grounded MCQs $\mathcal{Q}$ that this video contributes to TraceAV-Bench. We now walk through the three stages in turn.

Stage 1: from minutes to events. Directly exposing downstream agents to the entire stream of minute-level captions would force them to reason over dozens of nearly-redundant entries, since adjacent minutes typically belong to the same scene. We therefore collapse $\mathcal{M}$ into discrete event blocks via a sliding-window segmentation agent. At each minute t , the agent jointly inspects the running summary of the currently open event, the current minute m_t , and a two-minute look-ahead (m_{t+1}, m_{t+2}), and decides whether m_t continues the event or marks a boundary. The look-ahead is what distinguishes our segmentation from a purely causal scan. When narrative threads briefly interleave (a cutaway, a flashback, a parallel scene), the agent can confirm whether the interruption is incidental or persistent before committing to a boundary, which prevents premature fragmentation of long coherent events. The output $\mathcal{E}$ is a partition of the timeline into self-contained event blocks, each carrying its constituent captions, a fused summary, and the union of its entity occurrences.

Stage 2: from events to multi-hop trajectories. Operating now on the much shorter sequence $\mathcal{E}$, a trajectory proposal agent generates candidate multi-hop chains together with a per-chain question-direction hint that records the focus, answer cue, distractor cue, and answer-mode preference of the question the chain is intended to support. To guarantee that the resulting chains exercise genuine multi-hop reasoning, every candidate is screened by two structural conditions. First, its temporal span must exceed δ event blocks, ruling out trivial chains over consecutive scenes. Second, every pair of adjacent events in the chain must share at least one named entity, so that each hop is bridged by explicit coreference rather than by thematic coincidence; this entity-bridge requirement is central to enforcing that no individual event in the chain is alone sufficient to answer the question. The chains that survive these conditions form $\mathcal{T}$ and are passed to question generation.

Stage 3: from trajectories to MCQs. For each retained trajectory, a question generation agent produces a four-option MCQ following the three generation principles stated in the main paper. An anti-shortcut formulation that forces the question to be answerable only by direct observation of the trajectory, trajectory-grounded distractors drawn from other events of the same video, and stylistic uniformity across the four options. The agent additionally annotates every hop with a modality label in {audio, video, audio-video} and a candidate minute timestamp; the timestamp is then refined by mapping the cited evidence to its closest raw minute caption when it falls outside the event’s time range, which yields the fine-grained per-hop grounding used by both the quality-assurance stage and the downstream evaluation. Finally, the four answer slots are randomly permuted on a per-question basis so that the position of the correct option is uncorrelated with the generation order, removing any positional bias that models might inherit from its prompting template.

C Statistics of TraceAV-Bench

This section provides the detailed statistics of TraceAV-Bench. Table 4 reports the aggregate numbers under this three-axis view, and Table 5 breaks the question pool down further by sub-task.

Video corpus. The 578 videos in TraceAV-Bench amount to 339.5 hours of continuous audio-visual content, with an average length of 35.2 minutes and a broad range from just over 10 minutes to 2.3 hours. The visual quality of the collected videos is high, with 73.7% of videos at 720p or above

Table 5: Per-sub-task statistics of TraceAV-Bench. *Hops*: mean number of evidence hops per question. *Span*: mean temporal gap, in minutes, between the first and last evidence hop. *M-ch*: number of multi-choice items.

Dim.	Sub-task (Abbrev.)	#Q	%	Hops	Span (min)	M-ch
AVR	Information Retrieval (IR)	140	6.4	2.87	13.5	9
	Temporal Sequencing (TS)	97	4.4	3.99	15.2	14
	Entity Tracking (ET)	124	5.6	4.19	18.3	43
	Forward Causal Reasoning (FCR)	73	3.3	3.11	10.1	14
	Backward Causal Reasoning (BCR)	89	4.0	3.53	14.1	44
	Cross-Modality Matching (CMM)	85	3.9	3.80	19.3	22
	Spatiotemporal Localization (SL)	227	10.3	3.40	12.1	13
VR	Spatial Reasoning (SR)	165	7.5	3.38	14.6	0
	Visual Counting (VC)	226	10.3	4.24	14.4	11
AR	Speech Context (SC)	130	5.9	3.22	15.7	23
	Environmental Sound (ES)	88	4.0	3.41	11.8	22
	Background Music (BM)	131	6.0	3.68	17.4	30
MH	Visual-to-Audio Deception (V2A)	230	10.5	3.60	14.2	30
	Audio-to-Visual Deception (A2V)	229	10.4	4.00	15.9	25
	Temporal Splicing Fallacy (TSF)	166	7.5	4.23	19.8	52
Total		2,200	100.0	3.68	15.1	352

and a non-trivial share at 4K. The corpus deliberately spans documentaries, vlogs, sports, music performances, and other genres so that no topical shortcut transfers across the benchmark.

Question pool. The 2,200 questions are uniformly four-option multiple-choice items but are not uniformly single-choice: 16.0% (352 items) are multi-choice, where two to four options can be simultaneously correct. Question stems average 43.6 words and options average 22.1 words.

Reasoning trajectories. Every question is grounded in an explicit multi-hop trajectory, which is the central structural property that distinguishes TraceAV-Bench from prior audio-visual benchmarks. A trajectory comprises 3.68 evidence hops on average and stretches across 15.1 minutes of video, so the typical question requires the model to integrate evidence from roughly four moments separated by a quarter of an hour of footage.

Per-sub-task structure. Table 5 breaks the 2,200 items down by the fifteen sub-tasks defined in Appendix A. Hop counts and temporal spans vary in interpretable ways across tasks. Visual Counting (4.24 hops) and Entity Tracking (4.19 hops) demand the deepest reasoning chains, since both require accumulating evidence over many events, while Information Retrieval (2.87 hops) is structurally the shallowest because the answer hinges on a single key fact assembled from a few cues. Spans are widest for Temporal Splicing Fallacy (19.8 min) and Cross-Modality Matching (19.3 min), both of which fundamentally rely on relating widely separated events.

D Quality Assurance

Quality assurance yield. The 2,200 questions retained in TraceAV-Bench is the result of an aggressive multi-stage quality filter applied to a much larger pool of candidate items emitted by the agentic generation pipeline of Section B. Table 6 reports, for each sub-task, the number of items produced before the quality assurance is applied, the number that ultimately enter TraceAV-Bench, and the corresponding retention rate. Across the fifteen sub-tasks, the overall retention rate is 50.5%. Roughly half of the candidate items are dropped during quality assurance. The two tasks that survive at slightly higher rates (the localization, counting, and hallucination tasks) are precisely those whose answer space is the most structurally constrained, such as integer minutes, integer counts, so a larger fraction of generated candidates already complies with the rule-based portion of the filter and never reaches the more aggressive LLM-based audits.

Table 6: Quality assurance yield per sub-task. *Pre-QA*: number of candidate items produced by the agentic generation pipeline before any quality filter is applied. *Released*: number of items retained in TraceAV-Bench. *Dropped*: items removed by the quality assurance stage. *Kept*: retention rate, computed as Released / Pre-QA.

Dim.	Sub-task (Abbrev.)	Pre-QA	Released	Dropped	Kept (%)
AVR	Information Retrieval (IR)	280	140	140	50.0
	Temporal Sequencing (TS)	215	97	118	45.1
	Entity Tracking (ET)	265	124	141	46.8
	Forward Causal Reasoning (FCR)	165	73	92	44.2
	Backward Causal Reasoning (BCR)	200	89	111	44.5
	Cross-Modality Matching (CMM)	195	85	110	43.6
	Spatiotemporal Localization (SL)	400	227	173	56.8
VR	Spatial Reasoning (SR)	340	165	175	48.5
	Visual Counting (VC)	400	226	174	56.5
AR	Speech Context (SC)	270	130	140	48.1
	Environmental Sound (ES)	195	88	107	45.1
	Background Music (BM)	280	131	149	46.8
MH	Visual-to-Audio Deception (V2A)	405	230	175	56.8
	Audio-to-Visual Deception (A2V)	400	229	171	57.3
	Temporal Splicing Fallacy (TSF)	350	166	184	47.4
Total		4,360	2,200	2,160	50.5

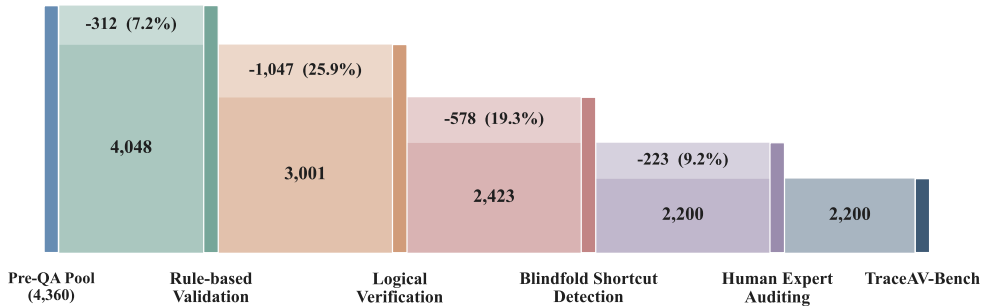


Figure 6: Flow of candidate items through the four-stage quality assurance pipeline.

Stage-wise attrition. Figure 6 visualizes how candidate items flow through the four sequential QA stages described in Section 3.2. Starting from a Pre-QA pool of 4,360 items, *Rule-based Validation* removes 312 format- or metadata-violating items, *Logical Verification* prunes the largest share (1,047 items) by filtering pseudo multi-hop questions and implausible distractors, *Blindfold Shortcut Detection* eliminates 578 items answerable without the video, and *Human Expert Auditing* discards a final 223 items through reviewer edits and batch rejection, yielding the 2,200 questions released in TraceAV-Bench.

E Additional Experimental Setup

All open-source models listed are deployed and evaluated on 8×NVIDIA H20 GPUs. We evaluate the closed-source models using their official APIs. We note that most current closed-source multimodal models, such as the GPT series, focus on the visual modality and do not natively accept audio input. Among the few that genuinely support joint audio-visual input, the Gemini series is the most representative. We therefore restrict our closed-source evaluation to the Gemini family. The specific versions used are as follows:

- Gemini-3.1-Pro-Preview: gemini-3.1-pro-preview

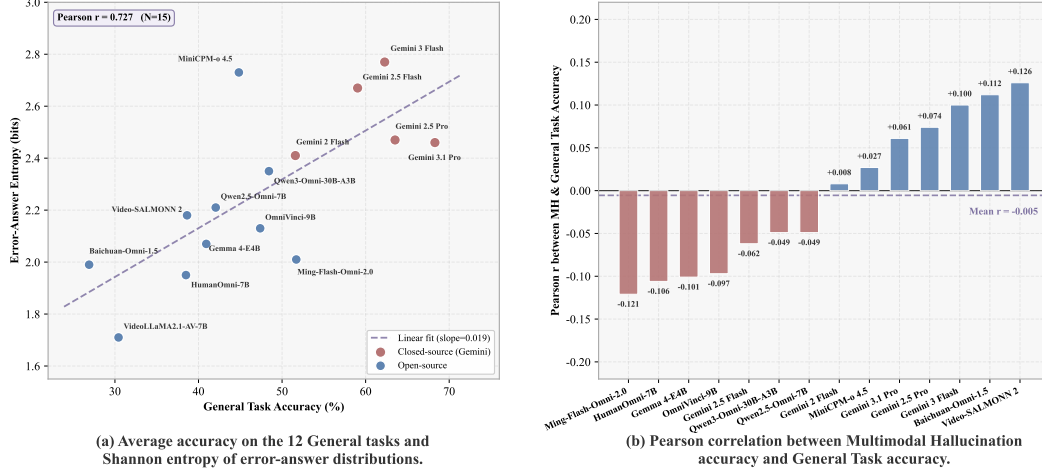


Figure 7: Two diagnostic analyses of OmniLLMs on TraceAV-Bench.

- Gemini-3-Flash-Preview: gemini-3-flash-preview
- Gemini-2.5-Pro: gemini-2.5-pro
- Gemini-2.5-Flash: gemini-2.5-flash
- Gemini-2.0-Flash: gemini-2.0-flash
- GPT-5.1: gpt-5.1-2025-11-13 (for data construction purpose only)

F More Experimental Findings

Stronger models make more *diverse* mistakes. Figure 7(a) plots each model’s general-task accuracy against the Shannon entropy of its error-answer distribution, $H = -\sum_{i=1}^k p_i \log p_i$, where p_i is the frequency of the i -th error pattern (the sorted tuple of predicted options) over incorrect predictions. The two are strongly correlated (Pearson $r=0.727$), with weak models such as VideoLLaMA2.1 (30.4%, 1.71 bits) collapsing errors onto a few patterns and strong models such as Gemini 3 Flash (62.3%, 2.77 bits) spreading them broadly. This tight coupling between capability and error diversity suggests that error-answer entropy may reflect a model’s potential, offering a indicator during evaluation throughout model training that complements what accuracy alone captures.

Video-level hallucination robustness and general-task ability are *nearly orthogonal*. Figure 7(b) shows, for each of the 14 models, the Pearson correlation between its Multimodal Hallucination (MH) accuracy and its general-task accuracy computed *across videos*. All 14 coefficients lie in the narrow band $[-0.121, +0.126]$, with mean $r=-0.005$, indicating that, at the video level, knowing how well a model comprehends a video provides little information on how robustly it resists deception on the same video. This echoes the model-level decoupling in Table 3, suggesting that, to deliver more trustworthy OmniLLMs, future training efforts should not focus solely on raw general-task accuracy but devote comparable attention to hallucination robustness.

G Prompt Templates

This section collects every LLM prompt used by the TraceAV-Bench pipeline, grouped by the stage in which it is invoked. Placeholder variables filled in at runtime are rendered as `<variable_name>`.

G.1 Pipeline Stage 1: Per-Minute Visual Captioning

The visual captioning stage processes each one-minute segment of a video while propagating a running entity cache to keep names and identities consistent across the entire video.

Stage 1: Visual Captioning with Entity Tracking

Task: Analyze this 1-minute video segment as a continuous narrative.

[Identity Consistency Context]
Refer to these previously identified entities to maintain naming consistency:
<entity_cache>

[Analysis Requirements]

- Narrative Flow: describe the events as a dynamic sequence, focusing on how actions evolve from start to finish (<minute_summary>).
- Entity Tracking: (1) identify which entities from the Context above are actively visible or interacting in this specific minute (<present_entities>); (2) if any known entity has changed appearance, state, or role, update its description in <present_entities> to reflect the latest state; (3) identify any NEW prominent people or objects not listed in the Context (<new_entities>).
- Detail Capture: capture subtle gestures, clothing details, and any on-screen text.
- Environmental Shift: note changes in setting, lighting, or atmosphere if applicable.
- For every entity in <present_entities> and <new_entities>, always return a non-empty distinguishing description.

[Strict Constraints]
DO NOT list any entities from the Context that are NOT visible in this segment.

[Output Format]
Your response must be a valid JSON object with the keys <minute_summary>, <present_entities>, and <new_entities>.

Figure 8: Prompt used in Stage 1 to generate per-minute visual captions while propagating an entity cache for cross-minute identity consistency.

G.2 Pipeline Stage 2: Audio-Visual Caption Fusion

The fusion stage takes the per-minute visual caption together with the corresponding raw audio track and produces a single audio-visual integrated caption while updating entity descriptions with newly observed audio evidence.

G.3 Pipeline Stage 3a: Event Segmentation

To convert minute-level captions into discrete events, we slide an LLM judge across the timeline. At minute T , the model is shown the running summary of the ongoing event, the current minute, and a two-minute look-ahead, and decides whether the current minute continues, transitions out of, or hard-cuts away from the active event.

G.4 Pipeline Stage 3b: Trajectory Proposal

Given a video’s event blocks, the trajectory proposer selects multi-hop chains of events that can support a high-quality question for the target task type τ . All fifteen task-specific prompts share the generic scaffold shown below. Note that only the task-specific *Core Principle* differs across tasks, and is listed individually for each task in Figures 13–27.

The fifteen task-specific instantiations (Core Principle for Stage 3b, Question Design and <task_specific_key> for Stage 3c) are listed individually in Figures 13–27 after the Stage 3c scaffold below.

Stage 2: Audio-Visual Caption Fusion

You are an expert video analyst. Your goal is to synthesize a 1-minute audio-visual integrated caption from two sources: (1) [Visual Context], a summary of what is seen, and (2) [Audio Track], the sounds and speech from the same segment.

[Visual Context]: `<visual_summary>`

[Visual Entities Present]: `<visual_entities>`

[Analysis Requirements]

1. Information Preservation: the integrated caption MUST encapsulate all information from the visual context.
2. Combined Narrative: integrate the visual description with the actual audio from this segment.
3. Entity Alignment and Evolution: match voices and sounds to the visual entities above when a logical connection exists. If the audio reveals new details about an entity (a spoken name, a distinct voice, a characteristic sound), update its description. If an entity has no relevant audio in this minute, omit it from `<updated_entities>`.
4. Audio Details: describe background music, environmental sounds, and the dialogue or speech context.
5. Final Output: a single, high-density paragraph that captures both what is seen and what is heard, maintaining chronological and identity consistency.

[Output Format]

A valid JSON object with keys `<minute>`, `<integrated_caption>`, and `<updated_entities>`.

Figure 9: Prompt used in Stage 2 to fuse the per-minute visual caption with the corresponding audio track into a single audio-visual caption.

Stage 3a: Event Boundary Segmentation

Task: Analyze the narrative continuity between a past event summary and the current video segment.

[Identity Consistency Context]

Active entities in the current minute: `<current_entities>`

[Past State Memory (Ongoing Event)]: `<past_summary>`

[Future Vision (Look-ahead Context)]: `T+1 = <future_1>`, `T+2 = <future_2>`

[Analysis Requirements]

Determine the relationship of the Current Minute (Minute T) to the Past State Memory:

- CONTINUE: direct continuation of narrative and characters.
- OVERLAP_TRANSITION: Minute T contains the resolution of the past and the start of the future.
- HARD_CUT: total break from the past, aligning instead with the new context in the Future Vision.

[Output Format]

A valid JSON object with keys `<action>`, `<reason>`, and `<updated_summary>`.

Figure 10: Prompt used in Stage 3a to slide an LLM judge across the timeline and decide CONTINUE / OVERLAP_TRANSITION / HARD_CUT at each minute, aggregating contiguous minutes into discrete events.

Stage 3b: Trajectory Candidate Proposer (Generic Scaffold)

Task: Select candidate multi-hop trajectories for `<task_name>` QA.

[Core Principle]
`<task_specific_core_principle>` (*per-task body in Figures 13-27*)

[Trajectory Quality]
Select only HIGH-QUALITY trajectories. A trajectory is high quality when ALL of the following hold:

1. STRICT Multi-hop: the intended question MUST be impossible to answer using any single event in the chain, and each hop must be a necessary piece of the puzzle.
2. No Common Sense / No Well-Known Facts: the answer MUST NOT be inferable from general world knowledge or widely known facts, and it must be uniquely tied to specific, non-obvious details of THIS video.
3. Challenging by design: the trajectory must support a question that would genuinely challenge an attentive viewer.
4. Strong evidence: every event in the chain contains concrete, specific evidence (named entity, exact number, quoted speech, visible action).
5. Clear entity bridge: adjacent events share at least one named entity whose state, role, or identity changes meaningfully between them.
6. Reject weak chains: do NOT include trajectories where the connection between events is only thematic or where the answer is guessable without watching the video.

[Question Direction]
For each candidate, propose a concrete and UNIQUE question direction. Across ALL candidates in this response, no two `<focus>` values may ask about the same fact, entity, or event. The fields `<focus>`, `<answer_hint>`, `<distractor_hint>`, and `<answer_mode_hint>` (single or multiple, respecting the per-task option-selection mode) must all be supplied.

[Output Format]
A valid JSON object containing a list `<trajectory_candidates>` of items, each with `<trajectory_id>`, `<event_ids>`, `<why_multihop>`, `<range_type>` (short or long), `<bridge_type>` (entity, semantic, temporal, or hybrid), and `<question_direction>`.

Figure 11: Generic scaffold for the Stage 3b trajectory proposer. The same scaffold is instantiated fifteen times, once per task type, by slotting in the task-specific Core Principle listed individually in Figures 13–27.

G.5 Pipeline Stage 3c: Multiple-Choice Question Generation

Given a selected trajectory, the question generator produces a four-option multiple-choice question, assigns each evidence step a modality label, and writes a per-event minute timestamp. The fifteen task-specific prompts again share a common scaffold. The variable parts are the *Question Design* block and the `<task_specific_key>` pair, both listed per task in Figures 13–27.

Per-Task Instantiations We now list, for each of the fifteen sub-tasks defined in Appendix A, the three pieces that instantiate the generic Stage 3b/3c scaffolds for that task. The *Core Principle* that fills the Stage 3b proposer (Figure 11), the *Question Design* body that fills the Stage 3c generator (Figure 12), and the `<task_specific_key>` pair that the Stage 3c output is required to emit. The fifteen figures are ordered to follow the same task taxonomy as Appendix A (AVR → VR → AR → MH).

Stage 3c: Multiple-Choice Question Generator (Generic Scaffold)

Task: Generate one multiple-choice `<task_name>` question.

[Question Design]

`<task_specific_question_design>` (*per-task body in Figures 13-27*)

[Question Quality Priority]

When generating the question, enforce ALL of the following:

- CHALLENGING: genuinely difficult for an attentive viewer; not obvious after a single viewing.
- NO Common Sense / NO Well-Known Facts: a person who has never seen the video must NOT be able to guess the answer.
- STRICT Multi-hop: removing even ONE evidence step must make the question unanswerable; every hop must be load-bearing.
- Strong grounding: every option must be grounded in trajectory evidence; no invented facts.
- Balanced options: all four options must be plausible on the surface.

[Modality Labeling Rule]

For each evidence step, assign `<label>` based on what is ACTUALLY USED to solve that hop: audio (only auditory cues), video (only visual cues), or audio-video (both modalities jointly required).

[Distractor Design Rules]

Distractors MUST satisfy ALL of: (1) factually grounded in entities, events, numbers, or phrases that actually appear in the video; (2) logically and commonsensically valid; (3) superficially plausible to someone who partially watched the video; (4) length parity with the correct option; (5) no giveaway language such as “none of the above”; (6) no surface-level tells in tone or grammar.

[Option Selection Constraint]

Receives a per-task constraint `<option_selection_constraint>`: if `single_choice_only`, set `<question_type>=“single”` with one item in `<correct_options>`; if `single_or_multiple`, either single-choice or multiple-choice is allowed (2 to 4 items in `<correct_options>` for the latter). Always keep exactly four options A/B/C/D.

[Timestamp Assignment]

For each node in `<final_trajectory>`, assign `<timestamp_minute>` by selecting the single minute within the event’s [`<start_minute>`, `<end_minute>`] range that best matches the evidence text, using the per-minute `<raw_segments>` for grounding.

[`<task_specific_key>`]

`<task_specific_key_pair>` (*per-task value in Figures 13-27*)

[Output Format]

A valid JSON object with keys `<final_trajectory>` (per-hop `<event_id>`, `<evidence>`, `<label>`, `<reason>`, `<timestamp_minute>`), `<task_specific_key>`, `<question_type>`, `<question>`, `<options>` (A/B/C/D), `<correct_options>`, and `<answer_text>`.

Figure 12: Generic scaffold for the Stage 3c question generator. The Question Design block and `<task_specific_key>` pair are slotted in per task type (Figures 13–27).

Task: Information Retrieval (IR)

[Stage 3b Core Principle]

Identify events where a specific key fact (on-screen text, number, spoken name, person identity) is embedded and can only be fully retrieved by chaining evidence from multiple events. Vary the answer position across the trajectory (beginning, middle, or end). Prefer events whose entities carry the attribute “audio-visual” so that both modalities contribute to retrieval. Include both short-range and long-range chains.

[Stage 3c Question Design]

Identify a specific key fact (on-screen text, number, spoken name, or person identity) as the retrieval target. The question must require chaining at least two evidence steps; the target fact must NOT be directly answerable from a single event. Vary where the answer sits in the chain. Distractors are plausible in-domain values (similar numbers, similar names) drawn from elsewhere in the trajectory.

[task_specific_key]

```
{<key_name>: retrieval_target_type, <key_value>: text | number | speech | person}
```

Figure 13: Per-task instantiation for **Information Retrieval (IR)**.

Task: Temporal Sequencing (TS)

[Stage 3b Core Principle]

Identify events that form a non-trivial temporal order or entity-state evolution that cannot be inferred from any single event. Prefer events where an entity’s appearance, role, or state visibly or audibly changes between events. Include both short-range and long-range chains. Avoid trajectories where the ordering is trivially recoverable from event IDs alone.

[Stage 3c Question Design]

Ask the model to order events, identify what happened before/after a given event, or describe how an entity’s state changed across the trajectory. The correct ordering must require evidence from all events. Distractors should be plausible alternative orderings or state sequences.

[task_specific_key]

```
{<key_name>: sequence_length, <key_value>: <number_of_ordered_items>}
```

Figure 14: Per-task instantiation for **Temporal Sequencing (TS)**.

Task: Entity Tracking (ET)

[Stage 3b Core Principle]

Centre each trajectory on a specific entity (person, object, or sound source) that appears in at least two non-adjacent events with meaningful state, role, or relationship changes. Leverage entities whose attribute is “audio-visual” when both modalities are needed to track the entity. Include both nearby transitions and far-apart isolated transitions. Ensure the question cannot be answered from a single event mention.

[Stage 3c Question Design]

Centre the question on a specific entity from the entity pool. Ask about the entity’s identity, role, relationship, or state at a specific point in the trajectory, requiring cross-event linking. Distractors should be other entities or plausible but wrong state descriptions drawn from the entity pool or trajectory context.

[task_specific_key]

```
{<key_name>: tracked_entity, <key_value>: <entity_name>}
```

Figure 15: Per-task instantiation for **Entity Tracking (ET)**.

Task: Forward Causal Reasoning (FCR)

[Stage 3b Core Principle]

Identify event pairs or chains where an earlier event is a clear prerequisite or cause for something that happens in a later event. The causal link must require evidence from both events; neither alone is sufficient to answer “what happens as a result of X”. Include both local causal chains and distant isolated causal chains.

[Stage 3c Question Design]

Frame the question as: “Given that [cause from early event], what happens / what is the outcome in [later event]?”. The causal link must be non-trivial and require evidence from both events. Distractors should be plausible but causally unrelated or reversed outcomes.

[task_specific_key]

```
{<key_name>: causal_direction, <key_value>: forward}
```

Figure 16: Per-task instantiation for **Forward Causal Reasoning (FCR)**.

Task: Backward Causal Reasoning (BCR)

[Stage 3b Core Principle]

Identify event pairs or chains where a later event shows an effect or outcome that can only be explained by tracing back to an earlier cause event. The backward causal link must require evidence from both events; neither alone is sufficient to answer “what caused X”. Include both local causal chains and distant isolated causal chains.

[Stage 3c Question Design]

Frame the question as: “Given that [effect / outcome in later event], what was the cause / prerequisite in [earlier event]?”. The backward causal link must be non-trivial and require evidence from both events. Distractors should be plausible but causally unrelated or forward-direction answers.

[task_specific_key]

```
{<key_name>: causal_direction, <key_value>: backward}
```

Figure 17: Per-task instantiation for **Backward Causal Reasoning (BCR)**.

Task: Cross-Modality Matching (CMM)

[Stage 3b Core Principle]

Identify events where a specific audio cue (speech phrase, sound, music) and a specific visual element co-occur or are explicitly linked across events. Prefer events containing entities with attribute “audio-visual”. Also identify events where an audio or visual element appears WITHOUT its expected counterpart, since these support plausible-but-wrong distractors. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask the model to correctly match an audio cue to its visual context (or vice versa) by chaining evidence across events. One option is the correct match; the other three are plausible mismatches (audio from a different event, visually similar but wrong scene). All evidence steps for this task type are labeled audio-video.

[task_specific_key]

```
{<key_name>: match_direction, <key_value>: audio->visual | visual->audio}
```

Figure 18: Per-task instantiation for **Cross-Modality Matching (CMM)**.

Task: Spatiotemporal Localization (SL)

[Stage 3b Core Principle]

Identify trajectories where pinpointing the EXACT MINUTE of a target event or entity state requires chaining clues from at least two events. Each event is provided with minute-annotated captions `<minute_captions>` keyed by absolute minute. Prefer multi-minute events so that fine-grained minute-level disambiguation is possible. The question's correct answer and all distractors MUST be concrete minute integers drawn from the trajectory's time range. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask the model to identify the EXACT MINUTE at which a specific event or entity state occurs, requiring evidence from at least two events to narrow it down. The question stem must describe the target event clearly enough that only one minute is correct. ALL FOUR options (A/B/C/D) MUST be concrete integer minute values drawn from the trajectory's time range, NOT descriptive phrases. The correct option is the minute confirmed by the trajectory evidence chain. Distractors should be other plausible minutes from the same trajectory.

[task_specific_key]

```
{<key_name>: target_minute, <key_value>: <correct_minute_integer>}
```

Figure 19: Per-task instantiation for **Spatiotemporal Localization (SL)**.

Task: Spatial Reasoning (SR)

[Stage 3b Core Principle]

Identify events where spatial relationships (position, direction, relative layout) of objects or persons can only be determined by combining visual evidence from multiple events. Prefer events with rich visual entity descriptions. Audio evidence should NOT be required to answer the question. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask about the spatial relationship (position, direction, relative layout) of objects or persons that can only be determined by combining visual evidence from multiple events. Audio evidence must NOT be needed. Distractors should be plausible but spatially incorrect alternatives. All evidence steps for this task type are labeled video.

[task_specific_key]

```
{<key_name>: spatial_relation_type,  
<key_value>: position | direction | layout | relative}
```

Figure 20: Per-task instantiation for **Spatial Reasoning (SR)**.

Task: Visual Counting (VC)

[Stage 3b Core Principle]

Identify trajectories where counting actions, state changes, or objects requires a condition established in one event and the counting target distributed across other events. Example pattern: “How many times does X do Y after Z happens?”. Visual evidence only; audio should not be required. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask how many times an action occurs, how many objects exist, or how many state changes happen, under a condition established in one event and counted across other events. Produce four distinct integer-count options. Distractors should be nearby integers ($\pm 1, \pm 2$) or counts derived from wrong conditions. Audio evidence must NOT be needed. All evidence steps for this task type are labeled video.

[task_specific_key]

```
{<key_name>: count_target, <key_value>: <what_is_being_counted>}
```

Figure 21: Per-task instantiation for **Visual Counting (VC)**.

Task: Speech Context (SC)

[Stage 3b Core Principle]

Identify events where spoken dialogue or narration across multiple events must be combined to answer a question about what was said. Prefer events containing entities with attribute “audio-visual” where the speech component is the primary evidence. Visual information should NOT be required. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask about what was said (spoken dialogue or narration) across multiple events, requiring the model to chain speech evidence from at least two events. Visual information must NOT be needed. Distractors should be plausible but incorrect speech content from the same trajectory or nearby events. All evidence steps for this task type are labeled audio.

[task_specific_key]

```
{<key_name>: speech_type, <key_value>: dialogue | narration | announcement | other}
```

Figure 22: Per-task instantiation for **Speech Context (SC)**.

Task: Environmental Sound (ES)

[Stage 3b Core Principle]

Identify events where non-human environmental sounds (rain, machinery, crowd, nature sounds) across multiple events must be combined to answer a question. The question must be answerable from audio alone without visual evidence. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask about non-human environmental sounds (rain, machinery, crowd, nature) that must be identified or reasoned about across multiple events. Visual information must NOT be needed. Distractors should be other plausible environmental sounds from the video. All evidence steps for this task type are labeled audio.

[task_specific_key]

```
{<key_name>: sound_category, <key_value>: <type_of_environmental_sound>}
```

Figure 23: Per-task instantiation for **Environmental Sound (ES)**.

Task: Background Music (BM)

[Stage 3b Core Principle]

Identify events where background music, ambient sound, or non-foreground audio cues across multiple events must be combined to answer a question. The question must be answerable from audio alone without visual evidence. Include both short-range and long-range chains.

[Stage 3c Question Design]

Ask about background music, ambient sound, or non-foreground audio cues that must be reasoned about across multiple events. Visual information must NOT be needed. Distractors should be other plausible background audio descriptions. All evidence steps for this task type are labeled audio.

[task_specific_key]

```
{<key_name>: background_audio_type, <key_value>: music | ambient | other}
```

Figure 24: Per-task instantiation for **Background Music (BM)**.

Task: Visual-to-Audio Deception (V2A)

[Stage 3b Core Principle]

Identify events with rich visual content (entities with attribute “visual” or “audio-visual”) where the visual scene strongly implies an audio detail that is NOT actually present in the video. The trajectory should provide enough visual evidence to make the hallucinated audio seem plausible, while the correct answer is that the audio does not exist. Include both short-range and long-range chains.

[Stage 3c Question Design]

Present real visual evidence from the trajectory and ask about an audio detail that does NOT actually exist in the video. The correct answer is the option that states the audio is absent or did not occur. The other three distractors should be fabricated but plausible audio details that the visual scene might suggest. The stem must NOT reveal that this is a hallucination test; phrase it as a genuine audio inquiry. Evidence steps are labeled video (visual cues drive the hallucination).

[task_specific_key]

```
{<key_name>: halluc_type, <key_value>: visual->audio}
```

Figure 25: Per-task instantiation for **Visual-to-Audio Deception (V2A)**.

G.6 Quality Assurance Prompts

After generation, every candidate item passes through two LLM-based quality checks: a *text-only solver* that flags items whose answer can be guessed without the video, and a *verifier* that audits multi-hop integrity, distractor quality, and answer leakage.

G.7 Evaluation Prompt

The same minimal prompt is issued to every candidate model on every TraceAV-Bench item, paired with the original video as input. We deliberately keep the wording short and free of chain-of-thought scaffolding. The model is asked to return only the letter (or comma-separated letters for multi-choice) of its selected option, which makes the response trivially parseable and avoids any prompt-side bias toward verbose explanations.

Task: Audio-to-Visual Deception (A2V)

[Stage 3b Core Principle]

Identify events with rich audio content (entities with attribute ‘audio-visual’) where the audio strongly implies a visual detail that is NOT actually present in the video. The trajectory should provide enough audio evidence to make the hallucinated visual seem plausible, while the correct answer is that the visual does not exist. Include both short-range and long-range chains.

[Stage 3c Question Design]

Present real audio evidence from the trajectory and ask about a visual detail that does NOT actually exist in the video. The correct answer is the option that states the visual is absent or did not occur. The other three distractors should be fabricated but plausible visual details that the audio might suggest. The stem must NOT reveal that this is a hallucination test; phrase it as a genuine visual inquiry. Evidence steps are labeled audio (audio cues drive the hallucination).

[task_specific_key]

```
{<key_name>: halluc_type, <key_value>: audio->visual}
```

Figure 26: Per-task instantiation for **Audio-to-Visual Deception (A2V)**.

Task: Temporal Splicing Fallacy (TSF)

[Stage 3b Core Principle]

Identify 2 to 4 events from DIFFERENT, non-adjacent time points whose real fragments could be falsely presented as a single coherent sequence. The trajectory should make the splice seem plausible (shared entities or similar settings) while the actual timeline makes the splice impossible. Prefer events with large temporal gaps between them.

[Stage 3c Question Design]

Present a fabricated narrative that splices real fragments from different, non-adjacent time points as if they form a coherent sequence. The correct answer must identify that the described sequence is impossible or did not happen as described. The other three distractors should accept the false narrative or propose alternative but equally false splices. Use the actual timestamps from the trajectory to ground the impossibility.

[task_specific_key]

```
{<key_name>: halluc_type, <key_value>: splice}
```

Figure 27: Per-task instantiation for **Temporal Splicing Fallacy (TSF)**.

QA-1: Text-only Direct Solver (System Prompt)

You are a strict multiple-choice solver. You only see question text and options. You do NOT have access to the video. You are NOT told whether this question is single-choice or multi-choice; infer from wording only. If confidence is insufficient, abstain instead of guessing. Return valid JSON only.

Figure 28: System prompt for the text-only solver, used to detect items whose answer can be guessed from textual cues alone.

QA-1: Text-only Direct Solver (User Prompt)

Solve the following benchmark question using text-only clues.
Question: <question>
Options: A. <option_A> B. <option_B> C. <option_C> D. <option_D>
Return a JSON object with <predicted_options> (a list of unique letters from A/B/C/D, possibly empty when abstaining), <confidence> (a number in $[0, 1]$), and <reason> (a brief justification).
Rules: do NOT assume single-choice by default; do NOT force a guess when uncertain; abstain by returning an empty list.

Figure 29: User prompt paired with the text-only solver system prompt in Figure 28.

QA-2: Item Verifier (System Prompt)

You are a strict QA benchmark verifier. Given the question, options, reference answer, and trajectory evidence, return JSON only. Evaluate quality and potential flaws conservatively.

Figure 30: System prompt for the item-level verifier that audits multi-hop integrity, distractor quality, and answer leakage.

QA-2: Item Verifier (User Prompt)

Verify this benchmark item.
Task type: <task_type> Question type: <question_type>
Question: <question>
Options: A/B/C/D = <options>
Reference correct_options: <correct_options> Reference answer_text: <answer_text>
Trajectory evidence: <trajectory_with_timestamps>
Please evaluate from the following angles:

1. Is the multi-hop chain real and correct?
2. Is this pseudo multi-hop (answer solvable by one hop or one local clue)?
3. Option quality: are there obvious superficial signals (e.g., one option much longer or stylistically different); are wrong options relevant rather than random noise; are wrong options plausibly confusing and semantically close?
4. Answer leakage: can the answer be directly inferred from wording in the question stem itself?

Return a JSON object with <is_real_multihop>, <multihop_issue>, <is_pseudo_multihop>, <pseudo_multihop_reason>, <option_quality> (with sub-fields <has_obvious_pattern>, <distractors_relevant>, <distractors_plausible>, <overall_quality> $\in$ {very_poor, poor, fair, good, excellent}, <reason>), <answer_leakage_detected>, <answer_leakage_reason>, <should_drop>, <drop_reasons>, <severity> $\in$ {none, low, medium, high}, and <summary>. The verifier sets <should_drop>=true for major quality flaws.

Figure 31: User prompt paired with the verifier system prompt in Figure 30; the structured JSON return is consumed by the dropping rule.

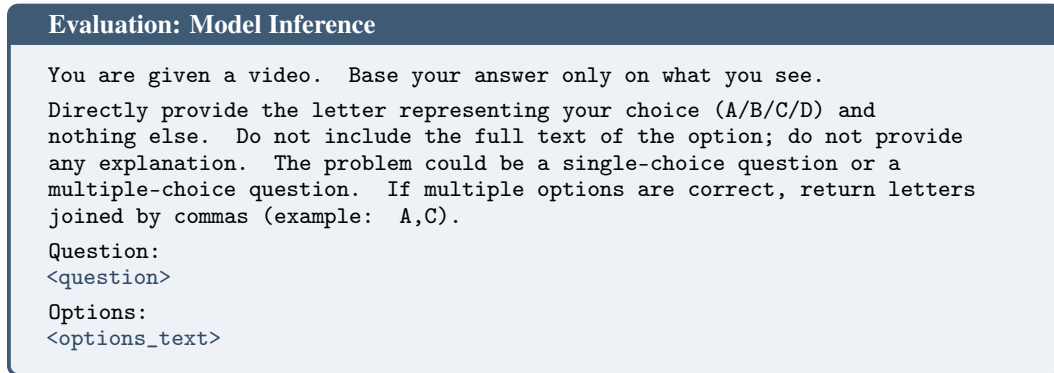


Figure 32: Unified evaluation prompt issued together with the source video to every candidate model on each TraceAV-Bench item.